

Result Analysis System

Node JS / React JS / Django (CS454PC)

Submitted

In partial fulfilment of the requirements for completion of

Bachelor of Technology

In

Computer Science and Engineering

By

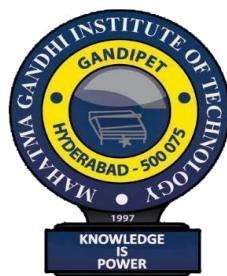
**Srinivas Konala
24261A05R0**

And

**Timothy Saxena
24261A05R4**

Under the guidance of

Ms. Anitha Talari
Assistant Professor
Department of Computer Science and Engineering

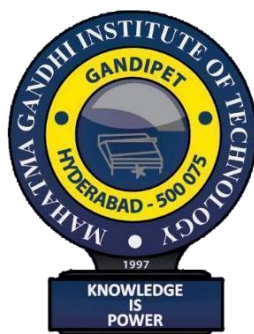


Department of Computer Science and Engineering,
MAHATMA GANDHI INSTITUTE OF TECHNOLOGY(A)
GANDIPET, HYDERABAD-500 075, INDIA.

2025-2026

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY(A)

(Affiliated to Jawaharlal Nehru Technological University Hyderabad)
Gandipet, Hyderabad-500 075, Telangana (India)



CERTIFICATE

This is to certify that the NODE JS/ REACT JS/ DJANGO (CS454PC) entitled **Result Analysis System**”, is being submitted by **Mr. Srinivas Konala** bearing **Roll No: 24261A05R0** and **Mr. Timothy Saxena** bearing **Roll No: 24261A05R4** in partial fulfilment for completion of **Bachelor of Technology IV Semester in Computer Science and Engineering** to **Mahatma Gandhi Institute of Technology** is a record of Bonafide work carried out by them under our guidance and supervision. The results embodied in this project have not been submitted to any other University or Institute for the award of any degree or diploma.

Project Guide
Ms. Anitha Talari
Asst. Professor, Dept. of CSE

Head of the Department
Dr. C.R.K Reddy
Professor, Dept. of CSE

Technology	Role	Where Used
Node.js	Server-side runtime	Backend
Express.js	HTTP framework & routing	Backend
MySQL	Relational database	Railway (cloud)
React + Vite	Frontend UI library & build tool	Frontend (Vercel)
JWT (jsonwebtoken)	Stateless authentication tokens	Backend + Frontend
bcryptjs	Password hashing	Backend
pdfkit	PDF marksheet generation	Backend
json2csv	CSV data export	Backend
xlsx	Excel file parsing	Import script
Vercel	Frontend static hosting + CDN	Cloud
Render	Backend Node.js hosting	Cloud
Railway	Managed MySQL hosting	Cloud

Table 3.1: Technologies and their roles in the project

3.6 bcryptjs

bcryptjs is used for one-way password hashing. Passwords are never stored in plain text. During import, each student's default password is hashed with bcrypt at a salt round of 10. At login, bcrypt's compare function tests the submitted password against the stored hash, returning true or false without ever reversing the hash.

3.7 pdfkit

pdfkit is a Node.js library for generating PDF documents programmatically without requiring a browser or headless Chrome. It is used to generate semester marksheets on demand. The backend constructs a complete A4 PDF in memory — including the college header, subject table, SGPA summary boxes, and a pass/fail status banner — then streams it to the browser as a downloadable file.

3.8 json2csv

json2csv converts JavaScript arrays of objects into CSV-formatted strings. It is used by the admin export routes to allow bulk download of filtered result data in spreadsheet-compatible format.

4. SOLUTION

4.1 System Architecture

The system follows a three-tier architecture. The presentation tier is a React application hosted on Vercel that communicates with the backend through HTTP API calls only. The application tier is a Node.js Express server hosted on Render that handles all business logic including authentication, GPA calculation, linear regression, and PDF generation. The data tier is a MySQL database hosted on Railway. No tier has direct knowledge of another tier's internal implementation, allowing each to be modified independently.

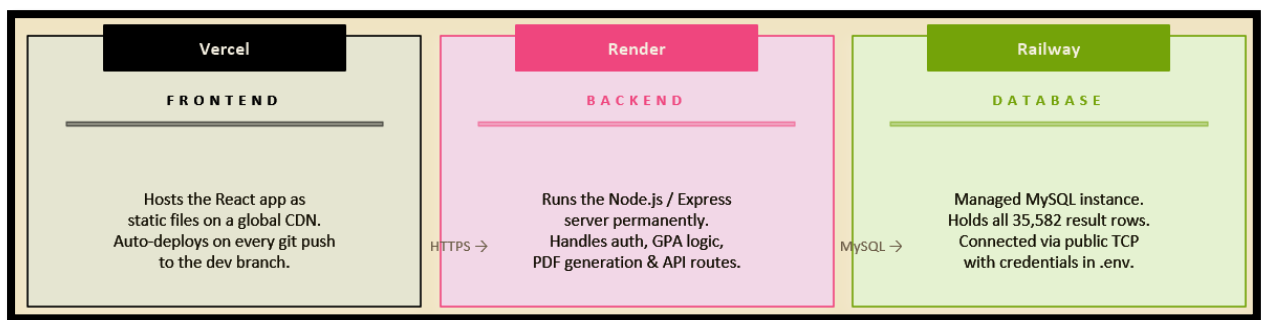


Figure 4.1: Three-Tier Architecture Diagram: Browser → Vercel (React) → Render (Express) → Railway (MySQL)

4.2 Database Design

The database consists of three tables. The students table stores one row per student with their hall ticket number as the primary key, a bcrypt password hash, and their assigned branch, year, and section. The results table is the core data store — each row represents a single student's performance in one subject in one semester. A unique constraint on (ht_no, semester, course_code) prevents duplicate entries and enables idempotent re-import. The faculty table stores accounts each scoped to a single course code. The admins table stores admin credentials with unrestricted access.

Table	Primary Key	Key Columns	Purpose
students	ht_no	password_hash, section, year, branch	1 row per student
results	id (auto)	ht_no (FK), semester, course_code, total_marks, grade_letter, credits	1 row per student-subject-semester
admins	id (auto)	username, password_hash	Admin accounts with full access

Table 4.1: Database schema overview

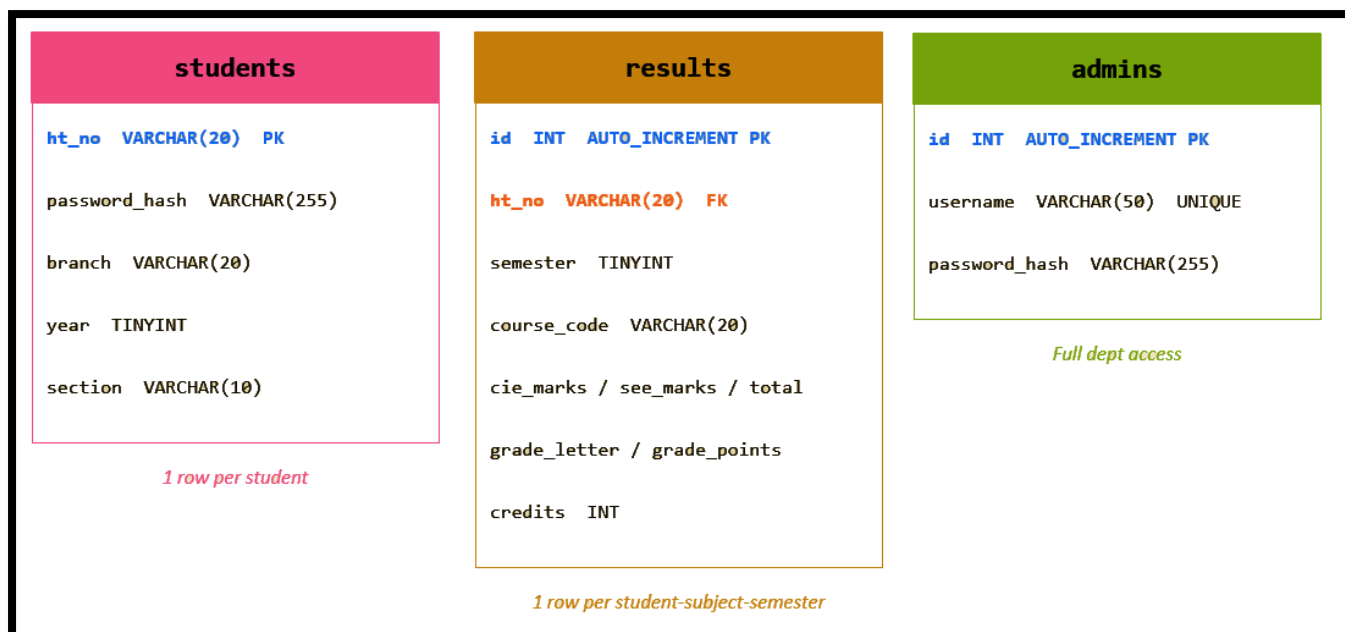


Figure 4.2: Entity-Relationship Diagram: students $\longleftrightarrow$ results, faculty, and admins tables with key columns and constraints

4.3 Excel-to-MySQL Data Pipeline

The department supplied result data as Excel files — one file per semester with approximately 3,000 rows each in the format: HT_NO, COURSE_CODE, COURSE_NAME, CIE_MARKS, SEE_MARKS, TOTAL_MARKS, GRADE_LETTER, GRADE_POINTS, CREDITS. A custom Node.js script (backend/scripts/import.js) was written to process these files and populate the database.

The script accepts two command-line arguments: the Excel file path and the semester number. The semester number is passed manually because it is not present in the Excel file itself. For each row, the script checks whether the student already exists; if not, it creates a new student record with a default password hashed with bcrypt. The result row is inserted using ON DUPLICATE KEY UPDATE, making the script safely idempotent. The semester number is passed manually because it is not present in the Excel file itself. For each row, the script checks whether the student already exists; if not, it creates a new student record with a default password hashed with bcrypt. Section assignment is inferred from the last two characters of the HT number using the department's roll range boundaries.

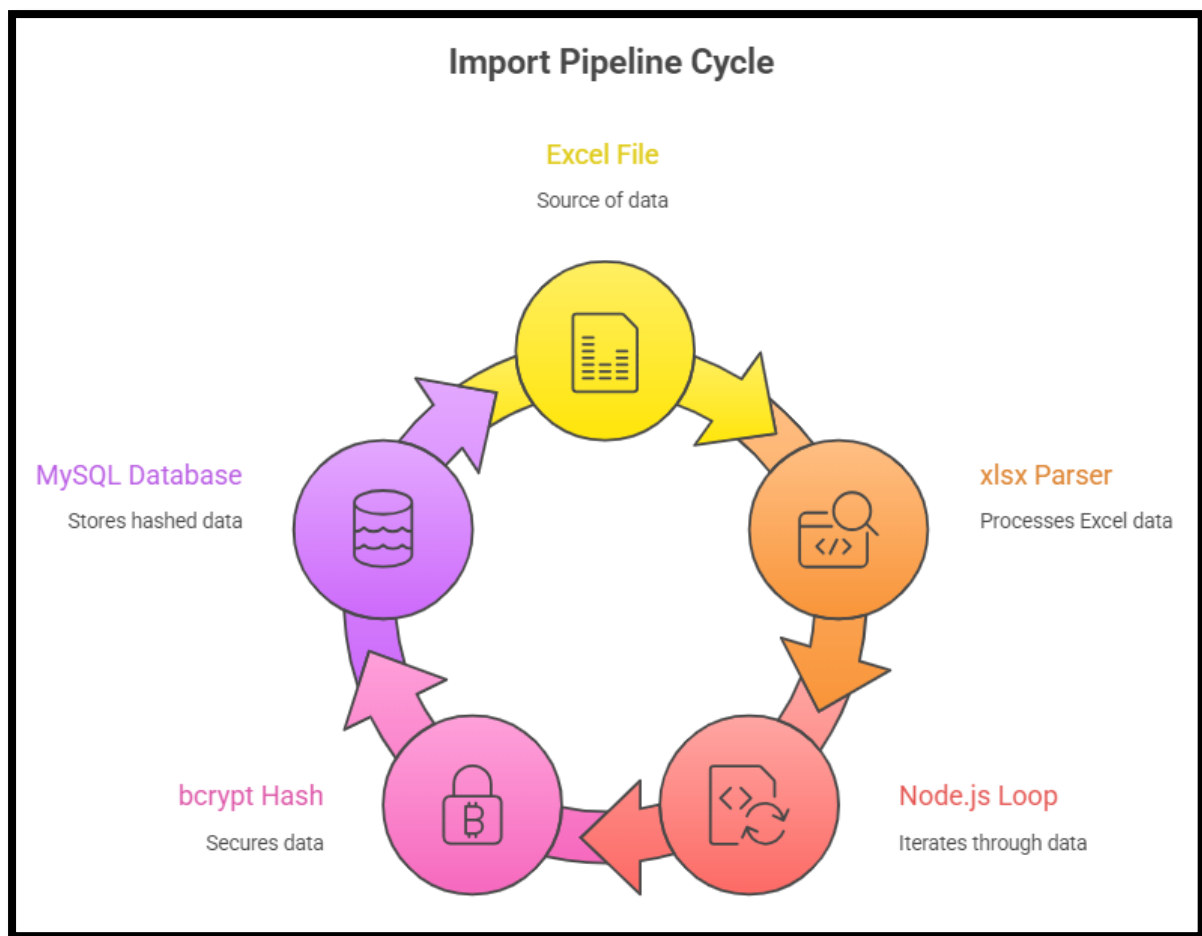


Figure 4.3: Import Pipeline Flow: Excel File → xlsx parser → Node.js loop → bcrypt hash → MySQL INSERT

4.4 API Route Design

The backend exposes RESTful endpoints grouped by role. Every protected route passes through a JWT verification middleware followed by a role guard.

Method	Endpoint	Role	Purpose
POST	/api/auth/student-login	Public	Student login, returns JWT
POST	/api/auth/admin-login	Public	Admin login, returns JWT
GET	/api/student/results	Student	Fetch all results, optional ?semester filter
GET	/api/student/sgpa	Student	SGPA per semester for trend chart
GET	/api/student/cgpa	Student	Overall CGPA and failed count
GET	/api/student/failed	Student	All failed subjects
GET	/api/student/rank	Student	Batch rank and percentile
GET	/api/student/marksheet/:semester	Student	Download PDF marksheet

GET	/api/admin/dashboard	Admin	Dept overview, section & semester tables
GET	/api/admin/toppers	Admin	Ranked list by CGPA or SGPA
GET	/api/admin/subject-analysis	Admin	Per-subject stats
GET	/api/admin/cgpa-distribution	Admin	CGPA histogram data
GET	/api/admin/section/:section	Admin	Drill-down for a specific section
GET	/api/admin/export	Admin	Full CSV export with optional filters

Table 4.2: Backend API route reference

4.5 Role-Based Access Control

Each role has a precisely defined data scope enforced at the server level. A student's JWT contains their HT number; every student route queries using `req.user.id`, making it impossible for one student to access another's records even if they manipulate the request. The admin role has no scope restriction and can query across all students, sections, and semesters.

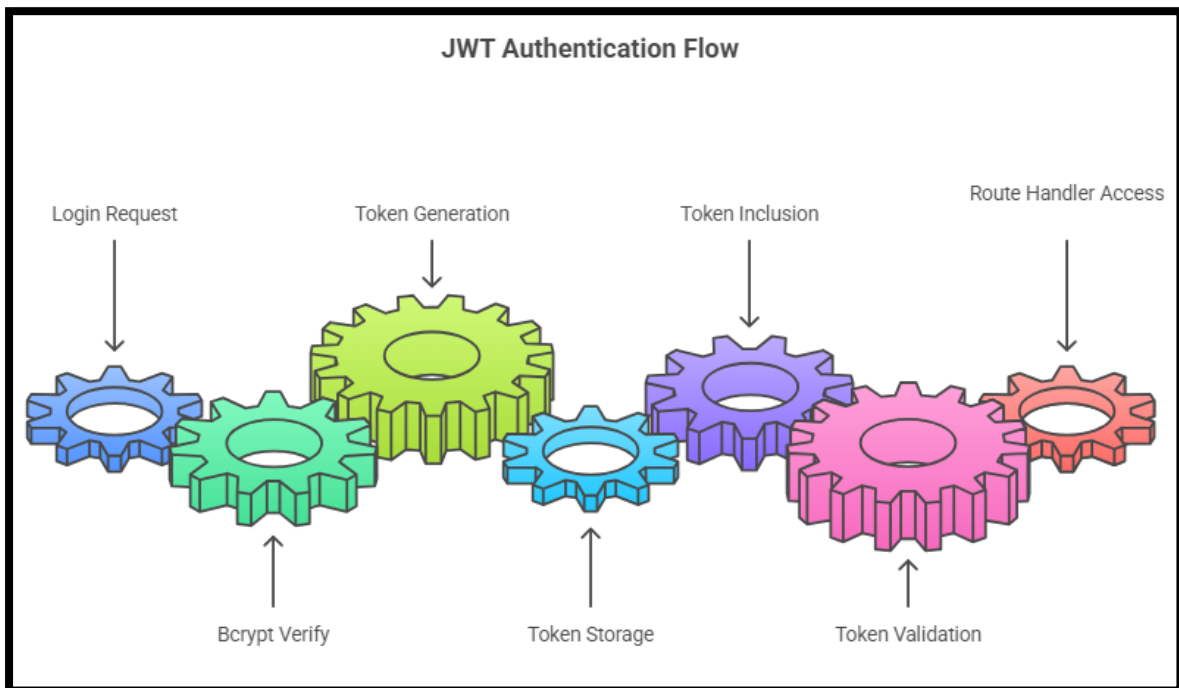


Figure 4.4: JWT Authentication Flow: Login Request → bcrypt verify → jwt.sign → Token stored → Bearer header → jwt.verify → Route handler

4.6 GPA Calculation

GPA calculation is implemented as a shared utility in backend/utills/gpa.js. The SGPA for a given set of results is the sum of (grade_points × credits) divided by total credits, excluding subjects with zero credits. CGPA applies the same formula across all semesters combined. Both student-facing routes and admin analytics use this single shared function, ensuring GPA values are consistent regardless of which part of the application computes them.

4.7 Linear Regression SGPA Projection

The GPA Overview tab includes a forward-looking SGPA projection for Year 3 and Year 4 students who have sufficient historical data. The projection is computed using simple linear regression applied to the student's existing SGPA values across semesters.

In linear regression, given a set of data points (x_i, y_i) where x is the semester number and y is the SGPA, the best-fit line is determined by calculating the slope (m) and intercept (b) as follows:

$$m = (n \times \Sigma(x \times y) - \Sigma x \times \Sigma y) / (n \times \Sigma x^2 - (\Sigma x)^2)$$
$$b = (\Sigma y - m \times \Sigma x) / n$$

Where n is the number of semesters completed. The projected SGPA for future semesters is then computed as: $\text{projected_SGPA} = m \times \text{semester_number} + b$. The projected values are capped between 0 and 10. The chart displays actual SGPA values as a solid line and projected values as a dashed line in a different colour, making the boundary between historical data and prediction.



Figure 4.5: Linear Regression SGPA Projection Chart: Actual SGPA (solid line) vs Projected SGPA (dashed line)

6. SCREENSHOTS

The following screenshots show the key screens of the Result Analysis System from the live deployment at <https://result-analysis-three.vercel.app>.

6.1 Login Page

The login page presents role tabs for Student and Admin. Each tab adjusts the identifier field label, placeholder, and the backend endpoint called on submission. On successful authentication the browser is redirected to the role-appropriate dashboard.

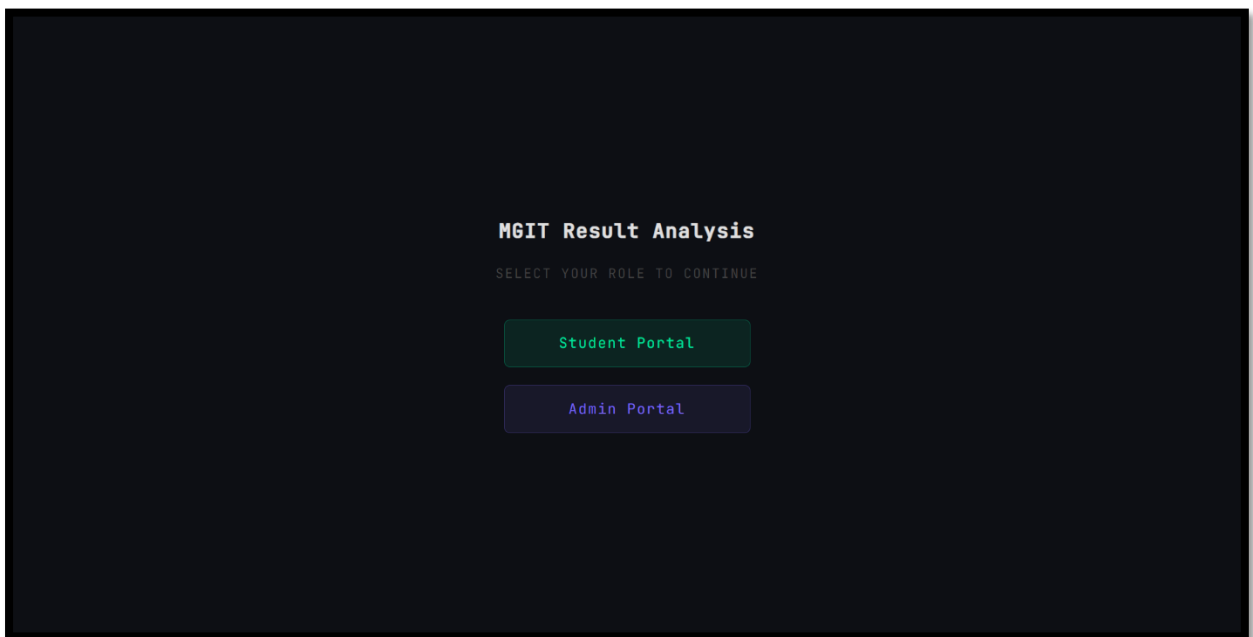


Figure 6.1: Login Page — Role tab switcher with Student and Admin tabs, identifier and password fields

6.2 Student Dashboard Tab

The Dashboard tab shows the MGIT logo, the student's HT number, four summary stat cards (CGPA, latest SGPA, semesters completed, backlog count), a batch rank card showing the student's rank and percentile within their batch, and a summary table of the most recent semester's results.

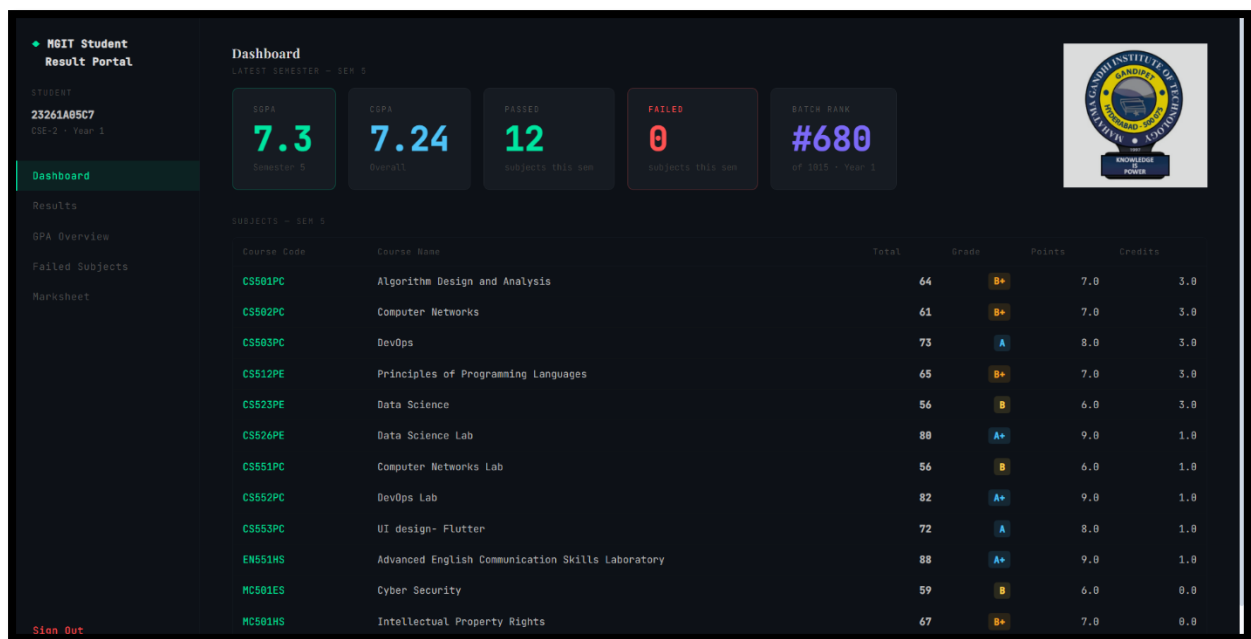


Figure 6.2: Student Dashboard Tab — Stat cards, batch rank card, and latest semester results table

6.3 Student Results Tab

The Results tab displays all subject results grouped by semester. A text search box and semester filter allow targeted browsing. Failed subjects are marked with a red indicator. Each semester group header shows the corresponding SGPA.

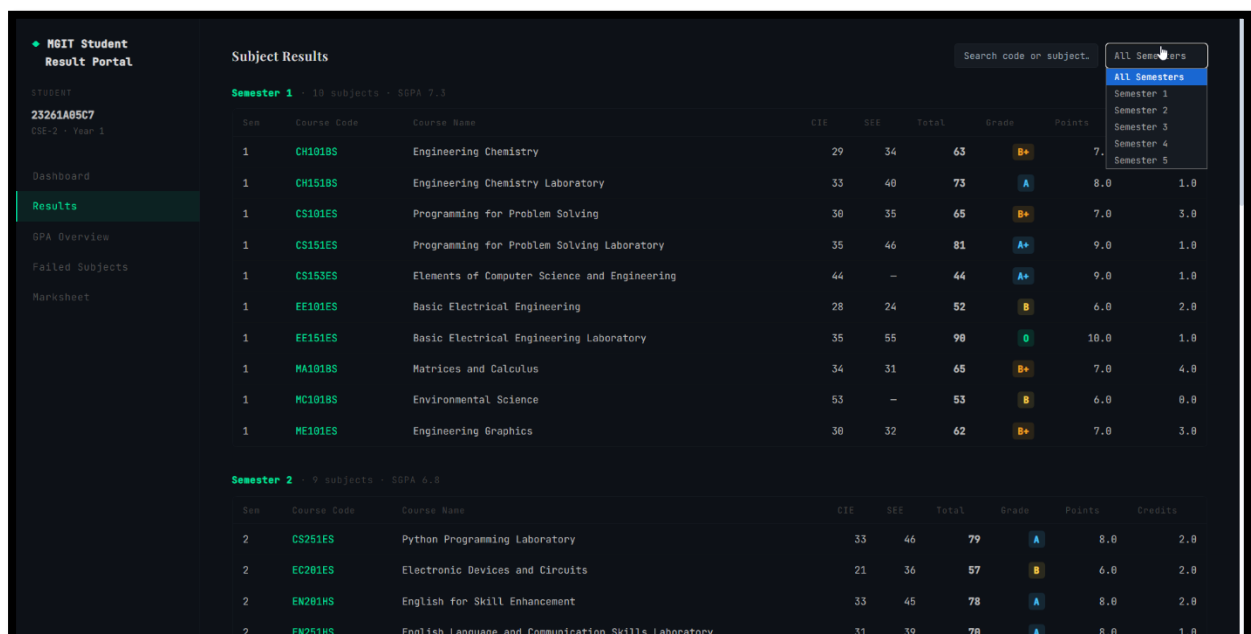


Figure 6.3: Student Results Tab — Semester-grouped table with search filter and grade badges

6.4 Student GPA Overview Tab — Actual Trend

The GPA Overview tab shows a cumulative CGPA trend line and a semester-wise SGPA chart. Both are rendered as pure SVG. The actual SGPA values are plotted as a solid line with labelled

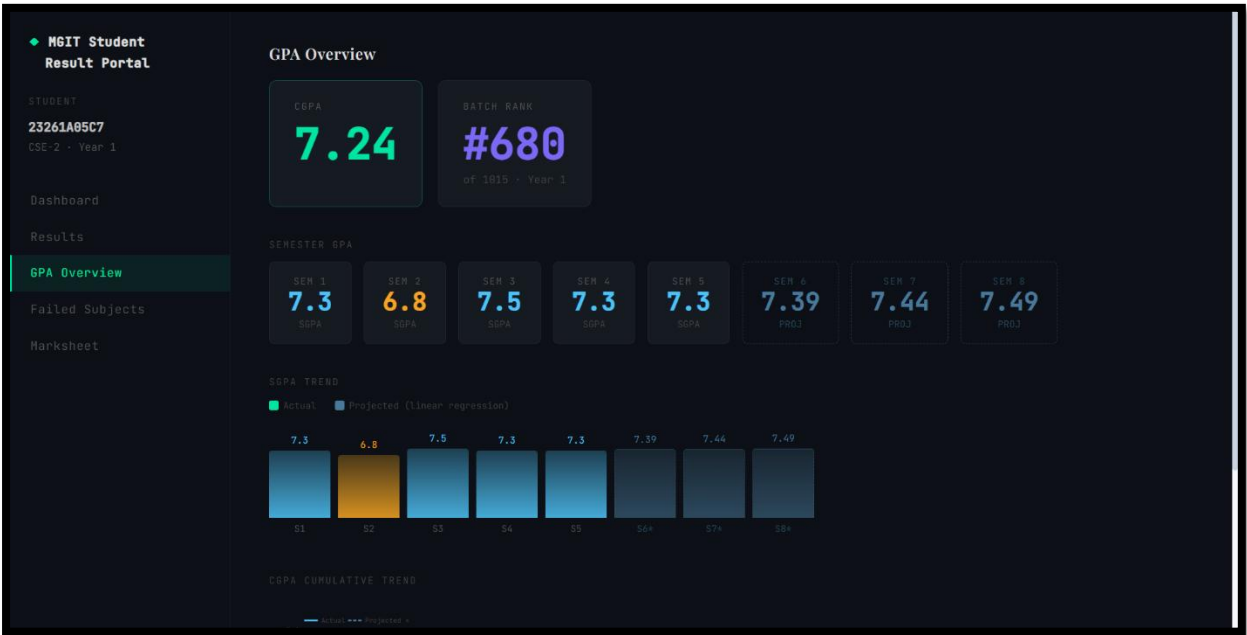


Figure 6.4: GPA Overview Tab — Cumulative CGPA trend and semester-wise SGPA chart (actual data)

6.5 Student GPA Overview Tab — Projection

For Year 3 and Year 4 students with sufficient historical data, the SGPA chart extends into future semesters using a linear regression projection. The projected portion is rendered as a dashed line in a distinct colour, clearly separating historical data from the statistical forecast.



Figure 6.5: GPA Overview Tab — Linear regression projection shown as dashed line beyond actual semesters

6.6 Student Failed Subjects Tab

The Failed Subjects tab lists all subjects in which the student has received an F grade. A semester filter narrows the view to a specific term. Each entry shows subject code, subject name, semester, marks obtained, and grade.

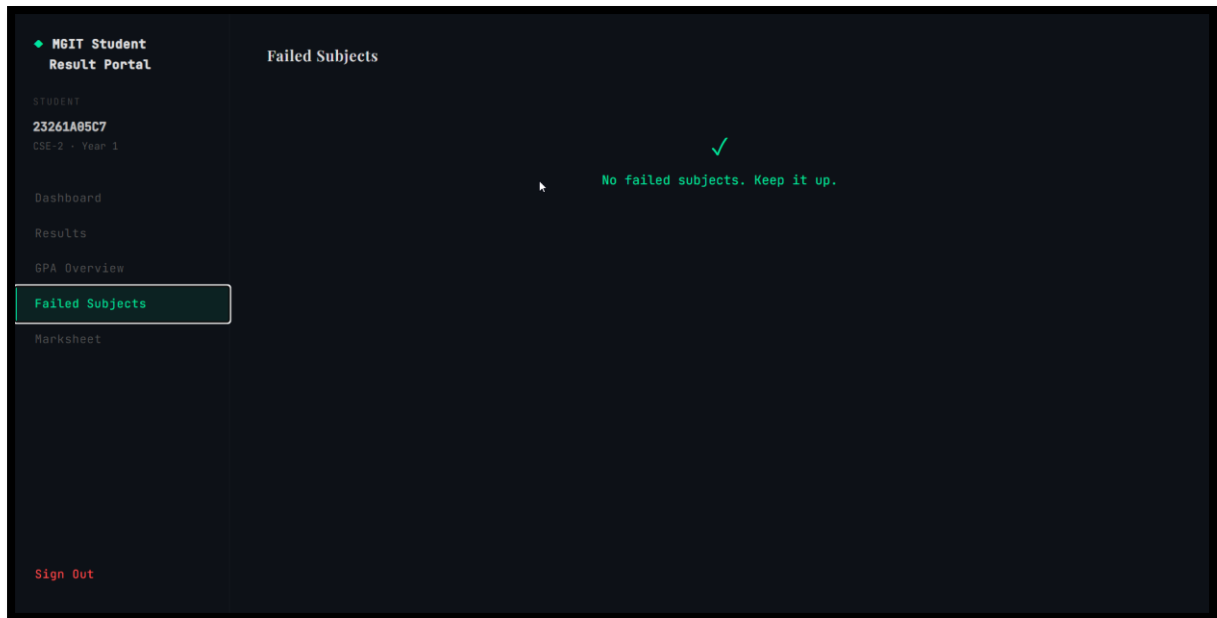


Figure 6.6: Failed Subjects Tab — Backlog list with per-semester filter

6.7 Student Marksheet Tab and PDF Output

The Marksheet tab shows one download card per semester with the SGPA for that term. Clicking download generates a formatted A4 PDF using pdftk. The PDF includes the college header, student details, a full subject table, an SGPA summary row, and a colour-coded pass/fail banner.

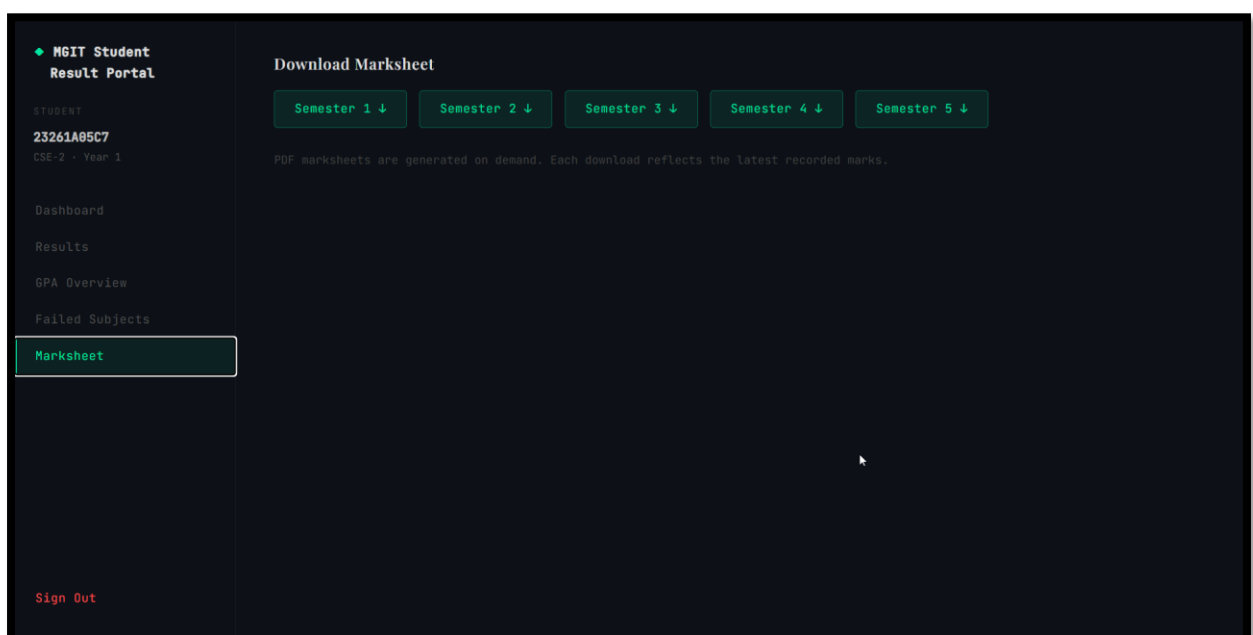


Figure 6.7: Marksheet Tab — Per-semester download cards

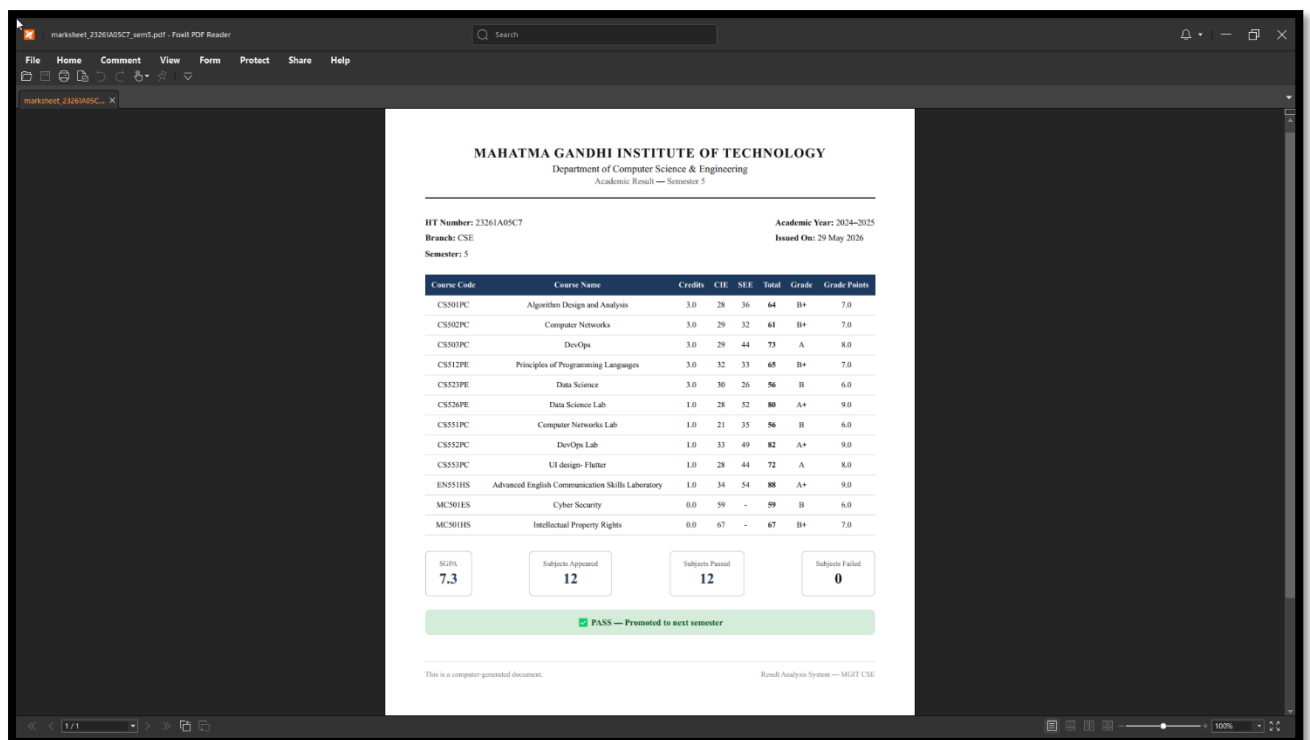


Figure 6.7.1: Generated PDF Marksheet — A4 layout with subject table, SGPA summary, and pass/fail banner

6.8 Admin Dashboard Overview

The Admin Dashboard tab shows four stat cards (total students, total results, total pass, total fail), a section breakdown table comparing CSE-1 through CSE-5, and a semester-wise performance table with average marks and pass percentages.

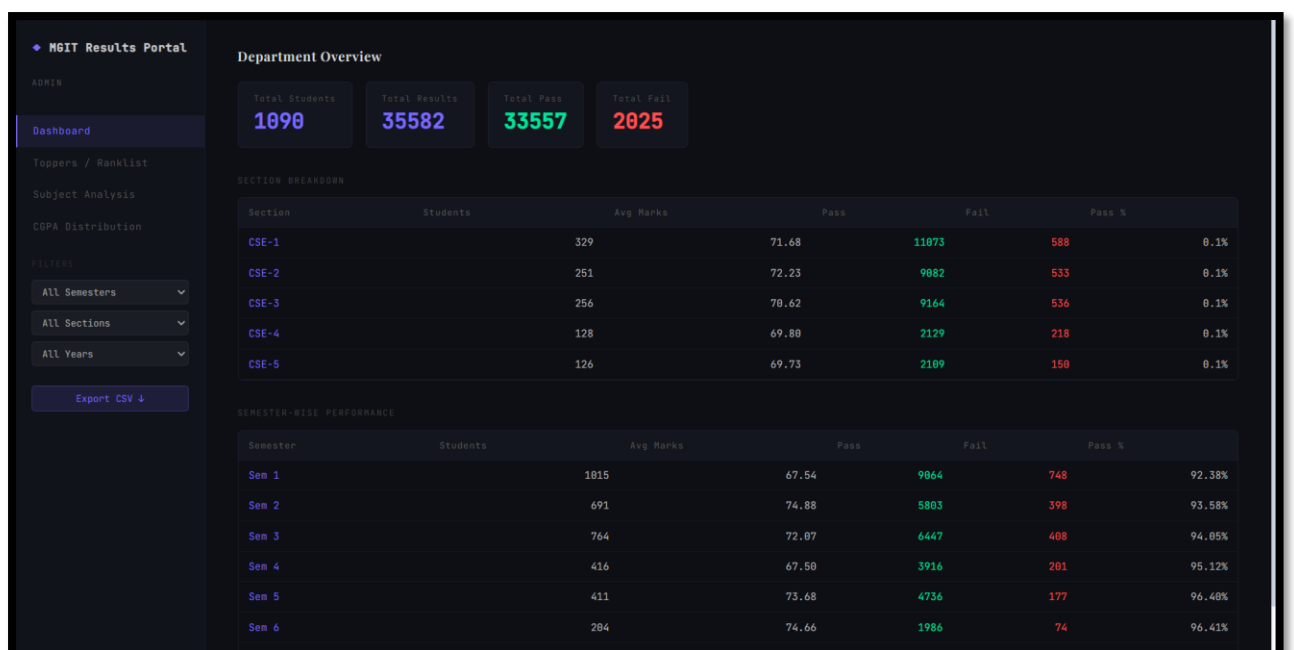


Figure 6.8: Admin Dashboard — Department stat cards, section breakdown, and semester performance table

6.9 Admin Toppers / Ranklist

The Toppers tab generates a ranked list by CGPA by default. Selecting a semester switches to SGPA-based ranking. Section and year filters allow cohort-specific drill-down. Tied GPA values receive the same rank number.

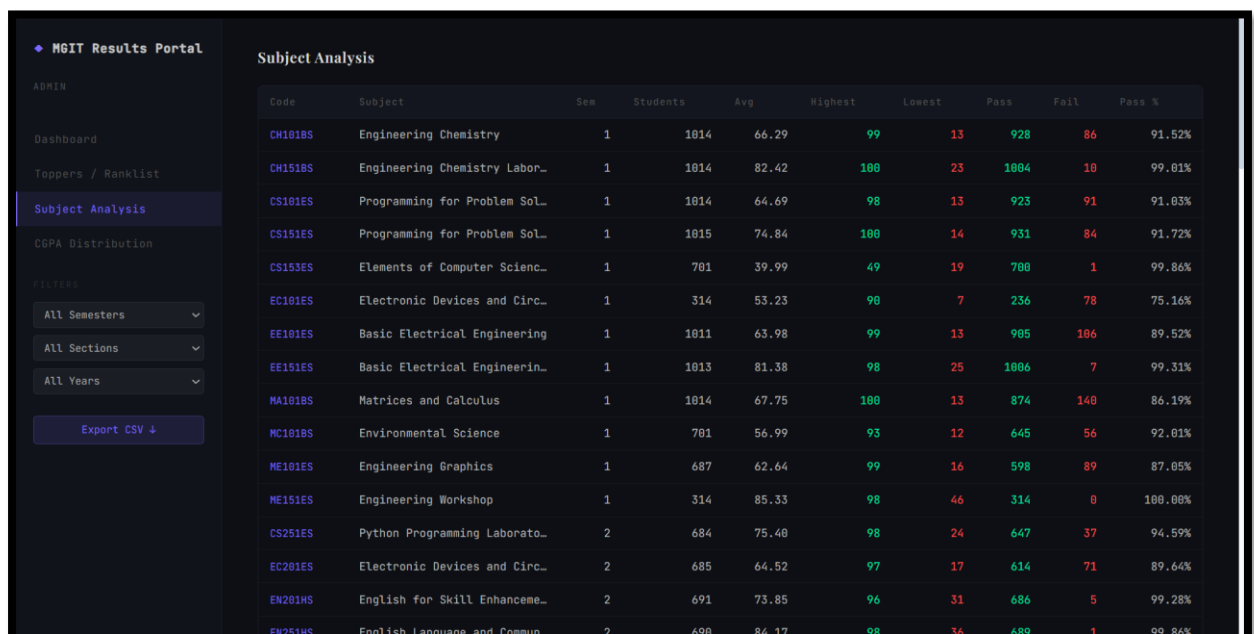


Rank	HT No	Section	Year	CGPA	Failed Subjects
#1	24261A0569	CSE-2	1	9.87	0
#2	25261A0551	CSE-1	1	9.85	0
#3	25261A05A2	CSE-2	1	9.70	0
#4	23261A0597	CSE-2	1	9.69	0
#5	24261A0580	CSE-2	1	9.65	0
#6	24261A0592	CSE-2	1	9.63	0
#6	24261A05U4	CSE-5	1	9.63	0
#8	24261A05L9	CSE-4	1	9.62	0
#9	24261A0595	CSE-2	1	9.60	0
#10	24261A05P9	CSE-4	1	9.57	0
#11	25261A0579	CSE-2	1	9.55	0
#11	25261A0555	CSE-1	1	9.55	0
#11	25261A0587	CSE-2	1	9.55	0
#14	23261A0553	CSE-1	1	9.53	0
#14	24261A05J8	CSE-3	1	9.53	0
#14	24261A05H4	CSE-3	1	9.52	0

Figure 6.9: Admin Toppers — CGPA-ranked student list with section, year, and semester filters

6.10 Admin Subject Analysis

The Subject Analysis tab shows a row per subject with average marks, highest and lowest scores, pass count, fail count, and pass percentage. A semester filter reduces the view to a specific term.



Code	Subject	Sem	Students	Avg	Highest	Lowest	Pass	Fail	Pass %
CH101BS	Engineering Chemistry	1	1014	66.29	99	13	928	86	91.52%
CH151BS	Engineering Chemistry Labor...	1	1014	82.42	100	23	1004	10	99.01%
CS101ES	Programming for Problem Sol...	1	1014	64.69	98	13	923	91	91.03%
CS151ES	Programming for Problem Sol...	1	1015	74.84	100	14	931	84	91.72%
CS153ES	Elements of Computer Scienc...	1	701	39.99	49	19	700	1	99.86%
EC101ES	Electronic Devices and Circ...	1	314	53.23	90	7	236	78	75.16%
EE101ES	Basic Electrical Engineering	1	1011	63.98	99	13	905	106	89.52%
EE151ES	Basic Electrical Engineerin...	1	1013	81.38	98	25	1006	7	99.31%
MA101BS	Matrices and Calculus	1	1014	67.75	100	13	874	140	86.19%
MC101BS	Environmental Science	1	701	56.99	93	12	645	56	92.01%
ME101ES	Engineering Graphics	1	687	62.64	99	16	598	89	87.05%
ME151ES	Engineering Workshop	1	314	85.33	98	46	314	0	100.00%
CS251ES	Python Programming Laborato...	2	684	75.40	98	24	647	37	94.59%
EC201ES	Electronic Devices and Circ...	2	685	64.52	97	17	614	71	89.64%
EN201HS	English for Skill Enhanceme...	2	691	73.85	96	31	686	5	99.28%
EN251HS	English Language and Commun...	2	690	84.17	98	34	680	1	99.84%

Figure 6.10: Admin Subject Analysis — Per-subject statistics table with pass percentage

6.11 Admin CGPA Distribution

The CGPA Distribution tab displays a horizontal bar chart with 0.5-step CGPA buckets. Summary cards below aggregate students into four broad performance bands with percentage share of the total cohort.

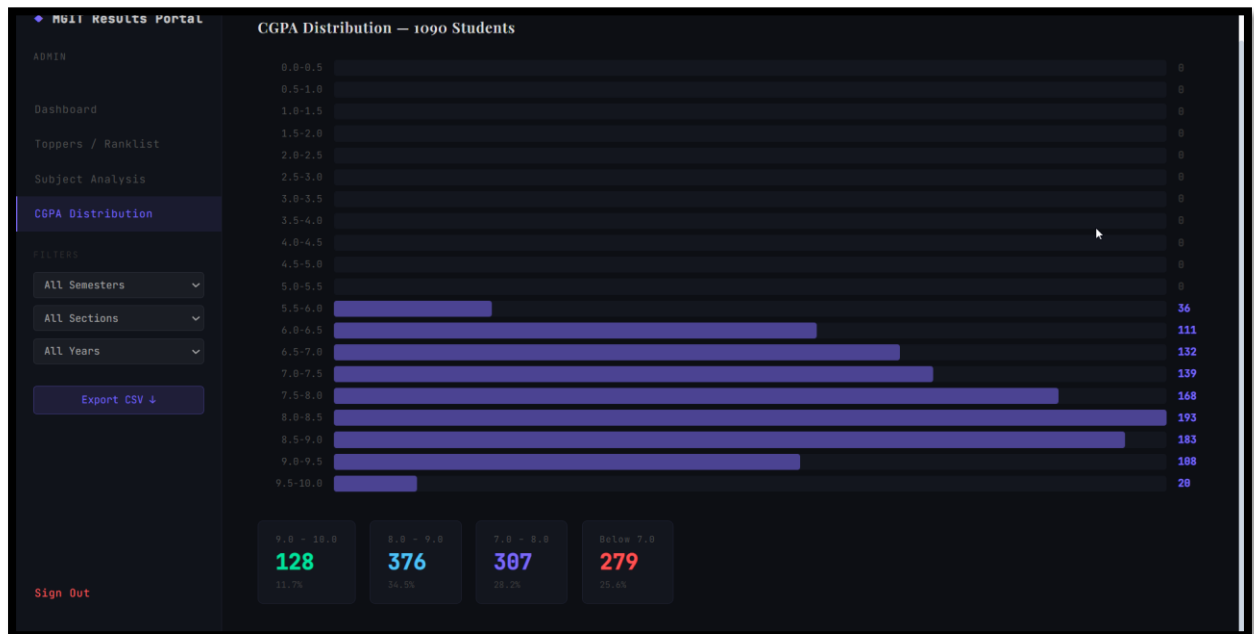


Figure 6.11: Admin CGPA Distribution — Histogram with 0.5-step buckets and summary band cards